# A cost effective smart insole system for real time gait analysis

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***Abstract*—Smart insoles have great potential as they offer real-time monitoring, analysis, and feedback for foot biomechanics, aiding in injury prevention and athletic performance improvement. Although off-the-shelf smart insoles exists, and solutions have been proposed, they are either very expensive, non-integrable, or lack sensors. In the current work, a low-cost fully integrated smart insole is introduced. The design integrates Force Sensing Resistors (FSRs), a 9-DOF IMU, Bluetooth Low Energy (BLE), and wireless charging. The system can transmit in real-time at 100Hz for more than 6 hours with a 40mAh battery. It can accurately identify walking and running patterns with great synchronization between the left and right foot.**

***Keywords— wearable; smart insoles; gait; force sensitive resistor; inertial measurement unit***

## I. Introduction

There has been a growing interest recently in developing smart insoles mainly due to their significant potential in protecting human health and enhancing performance. For example, in middle and old ages it prevails the ankle and foot pain, with the latter being a risk factor for impaired balance, locomotor disability, and functional activities of daily living [1]. Moreover, planus foot posture and pronated foot function are associated with arch, foot, and heel pain [2], while the rate of the repercussion of injuries and chronic conditions related to running is as high as 92% [3].

The omnipresence of gait analysis is evident from its wide-range applications in various disciplines, e.g., medicine, sports, genetic care, rehabilitation, and diagnostic studies [4]. Accurate gait analysis requires precise measurements. Motion labs, utilize high-tech equipment to acquire and analyze quantitative results, including 360 cameras, body-attached sensors, treadmills, and interconnected processing units, offering highly accurate results, in the sake of high cost.

The smart insoles eliminate these problems by integrating sensors and electronics to collect, store and transmit gait data. They are lightweight and gather quantitative data from patients, athletes or workers during their regular routine. The way the data are interpreted, and the processing algorithms make the smart insoles a valuable tool for many fields [5] such as: sports by preventing injuries and increasing the performance of the athletes [6], rehabilitation for monitoring the progress of the patients [7], medical diagnostics [8], first responders by increasing the situational awareness [9], Occupational Health and Safety (OHS) improvement for workers [10] and miners [11].

Several wearable sensor systems for quantitative gait analysis have been proposed, however some of them are not fully integrated [12, 13, 14, 15, 16] while others use only IMU sensors [17] or pressure sensors [18, 19]. In [20], an interesting smart insole design was proposed, but with low battery life span and transmission rate. The F-Scan [21] and Pedar System [22] exhibit excellent resolutions, yet their affordability is limited, and their primary usage revolves around clinical diagnostics and research objectives.

In this work, a cost effective fully integrated smart insole, named *Insofeet,* is proposed to detect gait patterns, which is fabricated with 3D printed materials and encapsulates FSR sensors, 9-DOF IMU and BLE communication. It provides quantitative gait analysis in real time with a 100Hz operation mode and a 40mAh battery with endurance of more than 6h.

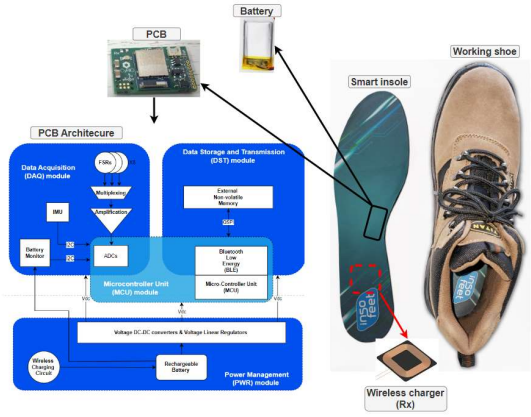


Figure 1: System overview

## II. Insofeet Smart Insole Architecture

In the following sections, the Insofeet architecture is broken down, focusing on the system's architecture, the sensors used, and the mechanical design and layers of the insole aiming to form a fully integrated design.

### A. Insofeet architecture

The proposed design is composed of (i) the Microcontroller Unit (MCU), (ii) the Data Acquisition (DAQ), (iii) the Data Storage and Transmission (DST), and (iv) the Power (PWR) modules.

The **MCU** is based on a SoC that supports BLE5 wireless connectivity. Serving as the system's core, the MCU establishes interconnections among all the modules. The **DAQ** module acquires data from the Force Resistive

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Sensors (FSRs) and the 9-DOF Inertial Measurement Unit (IMU) and incorporates a battery gauge IC to monitor battery parameters such as voltage level, current consumption, Stage of Charge (SoC), health, and aging. The **DST** module facilitates two-way communication with the host device and stores the raw data in non-volatile memory. The **PWR** module manages the system power requirements and includes the charging system, battery, and battery protection.

On the Printed Circuit Board (PCB), the 40mAh LiPo battery, pressure sensors, and wireless charging coil are interconnected. The battery and the PCB are positioned in the midfoot region, to minimize the load applied to the electronics. The wireless charging coil is located in the hindfoot region to facilitate charging, particularly when the insoles are inserted in shoes, eliminating the need for their removal by the user.

### *B. Sensors*

The Force Resistive Sensors (FSRs) demonstrate significant characteristics including high sensitivity, a wide dynamic range, fast response time, flexibility, a thin design, low power consumption, and cost-effectiveness. In the present design, Flexiforce sensors by Tekscan [23] were employed. The sensors system complements the 9-DOF IMU that integrates a 3-axis accelerometer, 3-axis gyroscope and 3-axis magnetometer.

### *C. Stack-up design*

The stack-up of the smart insoles consists of 3D-printed layers, each serving a specific purpose in the overall design and functionality. The thickness is 12mm in the middle of the plantar area and 6.5mm on the remaining surface. Layer 1 - Bottom cover: defines the positions of the integrated sensors responsible for collecting data on pressure distribution. Layer 2 - Enclosure Layer: securely houses the PCB and the battery. It provides protection against vertical and shear forces, ensuring the longevity and reliability of the internal components. Layer 3 - Top cover: seals the sensors and electronics, safeguarding them from external elements. It ensures the durability and stability of the integrated components. Layer 4 - Foam Layer: Attached on top, the foam-based layer provides comfort and support while wearing the insoles. Figure 1 depicts the system overview.

## III. SENSORS CHARACTERIZATION

To characterize the Force Resistive Sensors (FSRs), a non-inverting Op-Amp circuit [2] was implemented, due to its excellent linearity in voltage output corresponding to the applied force over the dynamic range. To enhance the dynamic range (Vref – Vsupply), a low reference voltage of 400mV was utilized. Furthermore, to prevent circuit saturation, the sensitivity was set to 80% output, for 100% loading, allowing for some overhead to accommodate force/pressure spikes. This approach ensures that the circuit does not saturate prematurely, even under maximum expected force conditions. Prior to characterization, a conditioning process was conducted, wherein the sensors were subjected to loads reaching 120% of the maximum expected force (set at 22kg, since this is the limit of the lineal region of the sensors) and cycled 3-5 times for 10 seconds each. Subsequently, a 3-point characterization was performed for all sensors to establish a correlation between the output voltage of the Op-Amp and known forces (in kg) applied using the digital force gauge DSV-1000N [24].

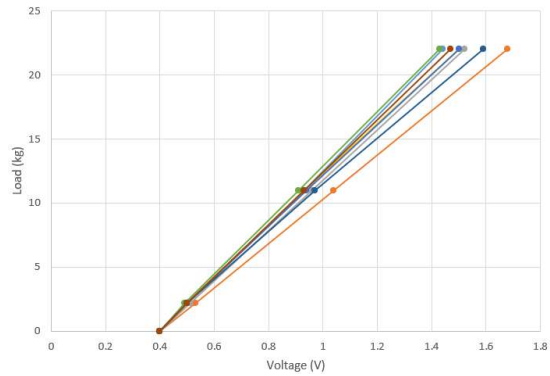


Figure 2: three-point characterization curves of sensors

Due to the inherent variation in pressure sensors from one unit to another, individual sensor characterization plays a critical role in obtaining accurate results. Figure 2 illustrates the deviations observed among the sensors, which arises from both human errors and the intrinsic variability inherent in the sensors themselves. The characterization factors for all sensors are determined using a linear equation.

## IV. TESTS & RESULTS

### *A. Battery lifespan*

The insoles were subjected to testing using a 40mAh LiPo battery at a sampling rate of 100Hz. Figure 3 illustrates the State of Charge (SoC) over time, depicting the battery entering the depleted region after 6H, 39 min, 29 seconds.

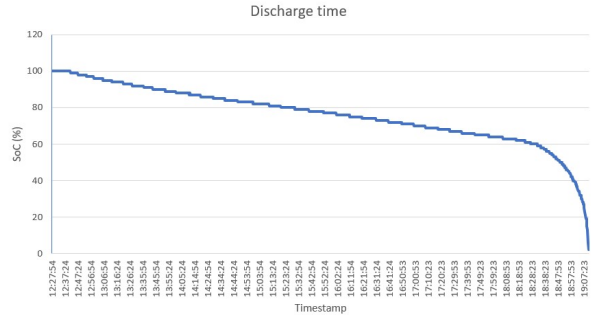


Figure 3: Battery discharge time

### *B. RSSI Measurements*

The primary consideration when incorporating the insoles into shoes, work shoes, or boots lies in ensuring a robust communication link between the Bluetooth devices. To assess this, measurements of the Received Signal Strength Indicator (RSSI) were conducted with the smart insoles installed in NO FLAME F2A boots utilized by firefighters. Four distinct use cases were examined, each corresponding to different ground morphologies. Real-time RSSI data was collected at a frequency of 1Hz. The scenarios of the tests performed are depicted in Figure 4-Figure 7, while Table 1 presents the average RSSIs for each case.

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TABLE 1: RSSI MEASUREMENTS IN DIFFERENT SCENARIOS

| Walking scenario | RSSI (dBm) |
|---|---|
| Clear of obstacles | -78.3 |
| Undergrowth | -80.5 |
| Shallow water | -89.1 |
| Mud covered | -81.5 |

Figure 4: Clear of obstacles

Figure 5: Undergrowth

Figure 6: Shallow water

Figure 7: Mud

The performance of the insoles was assessed through walking and running experiments, as depicted in Figure 8-10. The insoles were installed in NO FLAME F2A boots, specifically designed for first responders, with a sturdy construction capable of withstanding demanding conditions. A BLE receiver was used and a custom Generic Attribute Profile (GATT) profile was developed to facilitate bidirectional communication. The objective was to evaluate packet loss and analyze gait patterns within the defined use case.

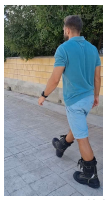
Figure 8: Walking

Figure 9: Running

All the data were collected with the verbal consent of the subjects. Figure 10 (top) presents the walking ground reaction forces observed in both the right and left feet. The graph illustrates that while one foot is in the swing phase (airtime) and the other foot is in the stance phase (ground reaction time). This alignment conforms to the expected walking pattern of the normal gait cycle [25]. It confirms the minimal data loss from both insoles and validates the effective distribution of sensors for accurately monitoring the complete gait cycle, encompassing the stance phases (heel strike, foot flat, midstance, heel off, toe off), as well as the swing phases. Furthermore, the data integrity is further corroborated by the running graph, which exhibits a peak corresponding to the heel strike of the opposite foot during each foot swing.

Figure 10: Walking (top) Running (bottom) GRF

To further investigate the packet loss 6 walking datasets were collected in 100Hz with duration between 110-114". In Table 2 are shown the received data packets from the left and right insoles with the respected deviation, while in

Table *3* shown the packet loss per insole.

TABLE 2: RIGHT/LEFT INSOLE PACKET DEVIATION

| Activity | Set# | Received packets | | Deviation |
|---|---|---|---|---|
| | | Left | Right | |
| Walking | Set1 | 11222 | 11421 | 1.74% |
| | Set2 | 10957 | 10945 | 0.11% |
| | Set3 | 11572 | 11570 | 0.02% |
| | Set4 | 11201 | 11205 | 0.04% |
| | Set5 | 11133 | 11144 | 0.10% |
| | Set6 | 11207 | 11270 | 0.56% |

TABLE 3: PACKET LOSS

| Activity | Set# | Packet Loss | | | |
|---|---|---|---|---|---|
| | | Left | Loss | Right | Loss |
| Walking | Set1 | 320 | 3% | 122 | 1.06% |
| | Set2 | 55 | 0% | 330 | 2.93% |
| | Set3 | 67 | 1% | 348 | 2.92% |
| | Set4 | 139 | 1% | 46 | 0.41% |
| | Set5 | 285 | 2% | 63 | 0.56% |
| | Set6 | 96 | 1% | 46 | 0.41% |

## V. CONCLUSIONS

In this work, a smart insole has been designed, fully integrated and field tested. The smart insoles were tested for power consumption, signal strength, and gait pattern detection. The results indicate a promising design for capturing body movement and foot pressure distribution data, transmitting those in real time at a high frequency. Considering that comfort is a key feature, future work will focus on optimizing the mechanical design and integration. Additionally, in remote areas where WiFi/5G connectivity is not available for transmitting information to remote hosts, alternative protocols such as LoRa, in combination with edge processing have research and industrial applicability value. This introduces the IoT Insofeet version, which aims to address connectivity challenges in those environments.

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